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11.3 Plant Water Relations

Water plays an essential role in the life of the plant. The availability of water influences the external and internal structures of plants as protoplasm is made of 60-80% water. Water is a **universal solvent** since most of the substances get dissolved in it and the high tensile strength of water molecule is helpful in the ascent of sap. Water maintains the internal temperature of the plant as well as the turgidity of the cell.

11.3.1 Imbibition

Colloidal systems such as **gum, starch, proteins, cellulose, agar, gelatin** when placed in water, will absorb a large volume of water and swell up. These substances are called **imbibants** and the phenomenon is **imbibition**.

Examples: 1. The swelling of dry seeds 2.The swelling of wooden windows, tables, doors due to high humidity during the rainy season.

The Power of Imbibition

In olden days, small wooden pegs were inserted into crevices of rocks followed by continuous hydration. Due to imbibition the volume of wooden peg increases and cuts off rocks precisely.

The gluten from wheat can take as much as 300% of its own weight

Significance of imbibition

i. During germination of seeds, imbibition increases the volume of seed enormously and leads to bursting of the seed coat.

ii. It helps in the absorption of water by roots at the initial level.

Activity

Imbibition experiment

Collect 5 gm of gum from Drumstick tree or Babool tree or Almond tree. Immerse in 100ml of water. After 24 hours observe the changes and discuss the results with your teacher.

11.3.2 Water Potential (Ψ)

The concept of water potential was introduced in 1960 by **Slatyer** and **Taylor**. Water potential is potential energy of water in a system compared to pure water when both temperature and pressure are kept the same. It is also a measure of how freely water molecules can move in a particular environment or system. Water potential is denoted by the Greek symbol **Ψ**

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